# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** A rope is 3.2 m long. After cutting off pieces of 45 cm, 55 cm and 60 cm, the remaining length is: 1

**(a)** 100 cm **(b)** 200 cm **(c)** 160 cm **(d)** 320 cm

**2.** A student heats the ends of four rods P, Q, R, S, each with a wax-fixed pin at the same distance. The pins fall at 10 s, 25 s, 40 s and 60 s. The best conductor is: 1

**(a)** S **(b)** Q **(c)** P **(d)** R

**3.** Thermometer A reads 0 °C in melting ice and 100 °C in boiling water. Thermometer B reads 2 °C and 98 °C. For the same hot water at actual 50 °C, B would most likely read: 1

**(a)** 54 °C **(b)** 48 °C **(c)** 52 °C **(d)** 50 °C

**4.** When ice at 0 °C is heated, the temperature stays at 0 °C till all ice melts. This shows that: 1

**(a)** ice cannot be heated **(b)** heat is used to change the state
**(c)** the heater is weak **(d)** the thermometer is faulty

**5.** In a pendulum experiment, 20 oscillations took 36 s. A second student gets 18 s for 10 oscillations. Both agree because: 1

**(a)** time period is 1.8 s in both **(b)** time period differs
**(c)** he is wrong **(d)** 36 s vs 18 s differ

**6.** A boy tests four materials A, B, C, D with a torch. Light passes fully through A, partly through B, not at all through C and D. Which can be used for a window to see out and still keep privacy? 1

**(a)** C **(b)** A **(c)** B **(d)** D

**7.** Rod A conducts heat twice as fast as rod B. If wax pins fall from A in 20 s from the heated end, the same pin on B would fall in about: 1

**(a)** 40 s **(b)** 10 s **(c)** 5 s **(d)** 20 s

**8.** Water of 100 g at 0 °C is converted to ice at 0 °C. The mass of ice formed is: 1

**(a)** 100 g **(b)** more than 100 g **(c)** 0 **(d)** less than 100 g

**9.** A student measuring a pencil places the 2 cm mark of the scale at one end and reads 12.5 cm at the other end. Length is: 1

**(a)** 10.5 cm **(b)** 2.0 cm **(c)** 14.5 cm **(d)** 12.5 cm

**10.** Why does a heavy ship made of steel float? 1

**(a)** its shape displaces much water **(b)** steel is lighter than water
**(c)** ships have no weight **(d)** water is sticky

**11.** The best conductor-insulator pair for a cooking pot and its handle is: 1

**(a)** glass and iron **(b)** wood and plastic
**(c)** plastic and aluminium **(d)** aluminium and plastic

**12.** A student claims that the water in a cup disappeared. The correct statement is: 1

**(a)** it vanished **(b)** it became oxygen
**(c)** it was destroyed **(d)** it changed to vapour and mixed with air

**13.** A road is 2.7 km. A runner has run 1850 m. The remaining distance is: 1

**(a)** 1850 m **(b)** 850 m **(c)** 750 m **(d)** 650 m

**14.** In the experiment above, the worst conductor is: 1

**(a)** P **(b)** Q **(c)** R **(d)** S

**15.** Hot water at 80 °C is poured into three cups: metal, ceramic and plastic. After 5 minutes the water is coolest in the: 1

**(a)** ceramic cup **(b)** plastic cup **(c)** metal cup **(d)** all equal

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** A road runs 3.6 km. A boy walks at a step of 60 cm. How many steps does he take? 2

**17.** Explain why an empty plastic bottle floats but a bottle filled with sand sinks. 2

**18.** Two beakers contain 100 mL water at 70 °C and 700 mL water at 70 °C. Compare their temperatures and say if the bigger one has more heat. 2

**19.** Explain why a wet cloth kept in a closed polythene bag does not dry. 2

**20.** Convert 0.003 km into cm. 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** The circumference of a wheel is 2.2 m. How many turns will it make to cover 1.1 km? 3

**22.** Substance X is a shiny solid that does not dissolve in water, sinks in water and becomes hot at the other end when one end is heated. Name its class of material and give two reasons. 3

**23.** A student measures the temperature of boiling water as 98 °C in a hill station and 100 °C in a plain. Suggest a reason. 3

**24.** A tank of 2000 L loses 4 L per day by evaporation. If it is filled once a month (30 days) how much is lost, and what percentage? 3

OR

Describe an experiment to measure the length of a curved line of 1 metre length marked on a floor by two students. 3

**25.** Explain the importance of non-conducting handles in electrical tools. 3

OR

Describe why the lid of a tiffin box should be airtight to keep food warm. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** A student wants to find the circumference of a tree trunk, the thickness of a coin and the length of a pencil. Choose the instruments and explain the method for each. **5**

**OR**

Design an investigation to rank wood, plastic, steel and copper for suitability as a cooking spoon handle. Name the variables and the safety steps. **5**

**27.** Describe an experiment to find which of three liquids (water, oil, milk) cools fastest. Explain controls, measurements and how to present results. **5**

**OR**

Design an experiment to find which of the three places (sunny, shady, windy) is best for drying clothes and how to compare. **5**

**END OF QUESTION PAPER** — **Total: 50 Marks**

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